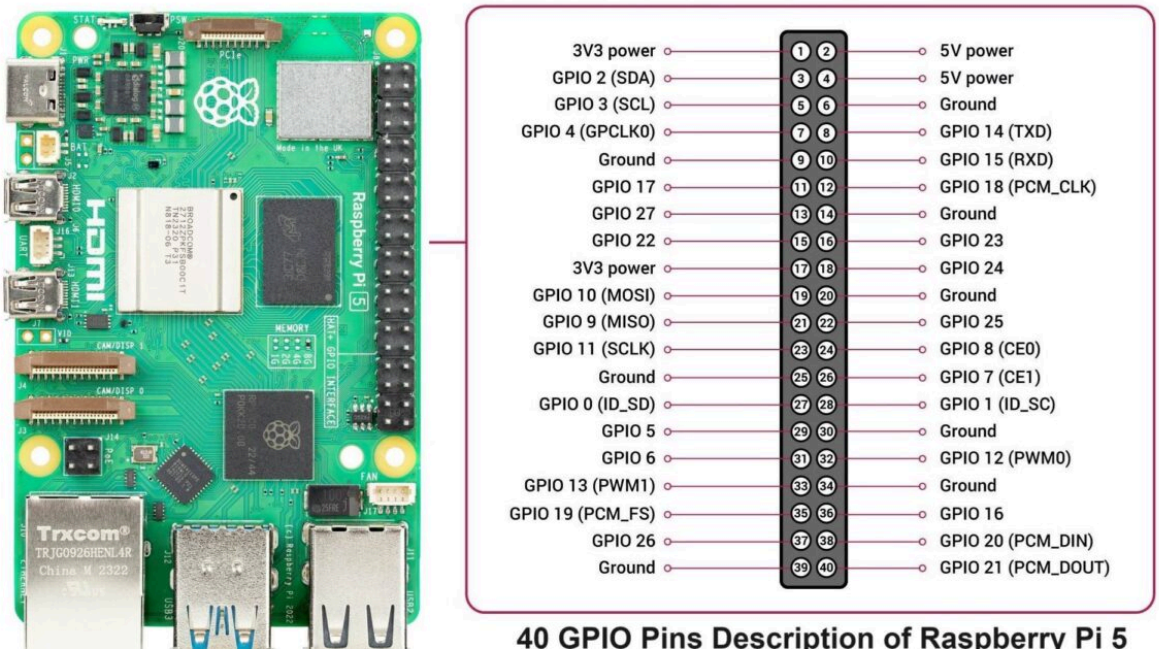
Introduction

1. Network credentials: `Laboratorium-IoT /`
2. Github repository - github.com/tocet/prog_devices



40 GPIO Pins Description of Raspberry Pi 5

Ex 1. Introduction to plotting - `pd_ex03_plot.zip`. Required libraries:

- *matplotlib* - matplotlib.org
- *seaborn* - seaborn.pydata.org
- *bokeh* - bokeh.org
- *tensorflow*

Ex 2. How to work with files - `pd_ex03_files.zip`.

Ex 3. How to work with the BMP-280 - `bmp280.py`.

Task. Application with a graphical user interface for displaying data from BMP-280:

- (0.1 points) use the BMP-280 to collect temperature and pressure data;
- (0.1 points) displaying data on a chart/charts;
- (0.1 points) all descriptions on the interface should be configurable in the JSON configuration file;
- (0.1 points) ability to load axis descriptions, units, chart title, and data from a JSON file;
- (0.1 points) add a grid on/off button;
- (0.2 points) add buttons to move a marker in the chart;
- (0.1 points) display marker coordinates;
- (0.1 points) display marker coordinates on a chart next to the marker;
- (0.1 points) button - save data to a CSV file.

For those interested:

1. Benefits of Bokeh over Python visualization libraries like Seaborn, Matplotlib & Plotly:
   holypython.com/benefits-of-bokeh-over-python-visualization-libraries-like-seaborn-matplotlib-plotly-2/?utm_content=cmp-true

2. Introducing Plotly AI:
   plotly.com

3. Data Visualization in Python with matplotlib, Seaborn, and Bokeh:
   machinelearningmastery.com/data-visualization-in-python-with-matplotlib-seaborn-and-bokeh/